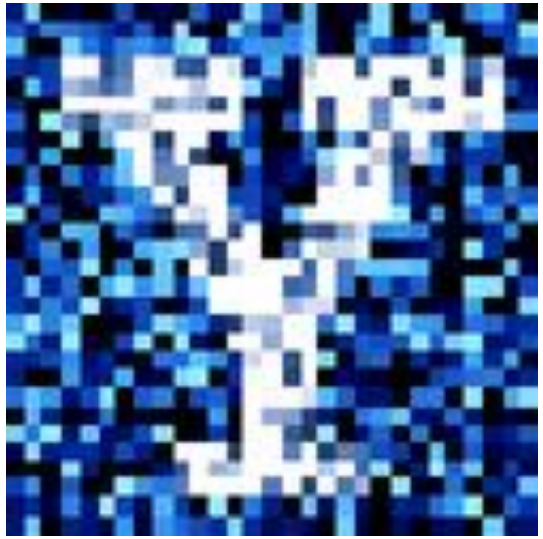
YData: Introduction to Data Science



Class 16: For loops and writing functions

Overview

Your data visualization!

Quick review of for loops

Conditional statements

Writing functions

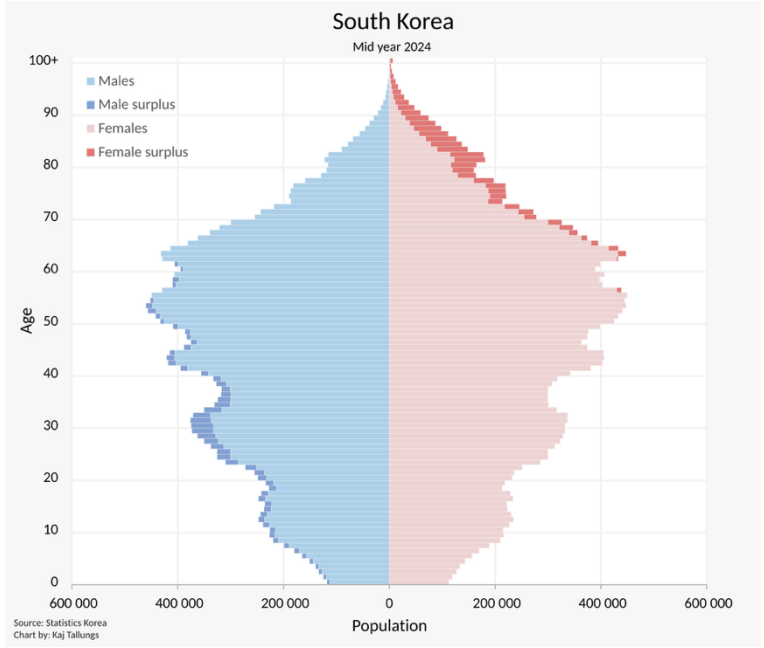
Reminder: start on your class project

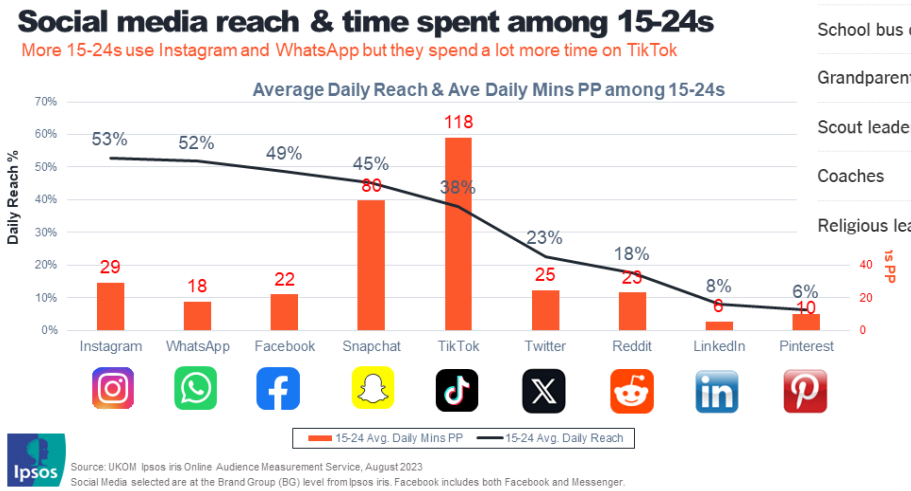
It'll be a good idea to start thinking about your project: a polished draft of the project is due on April 10th

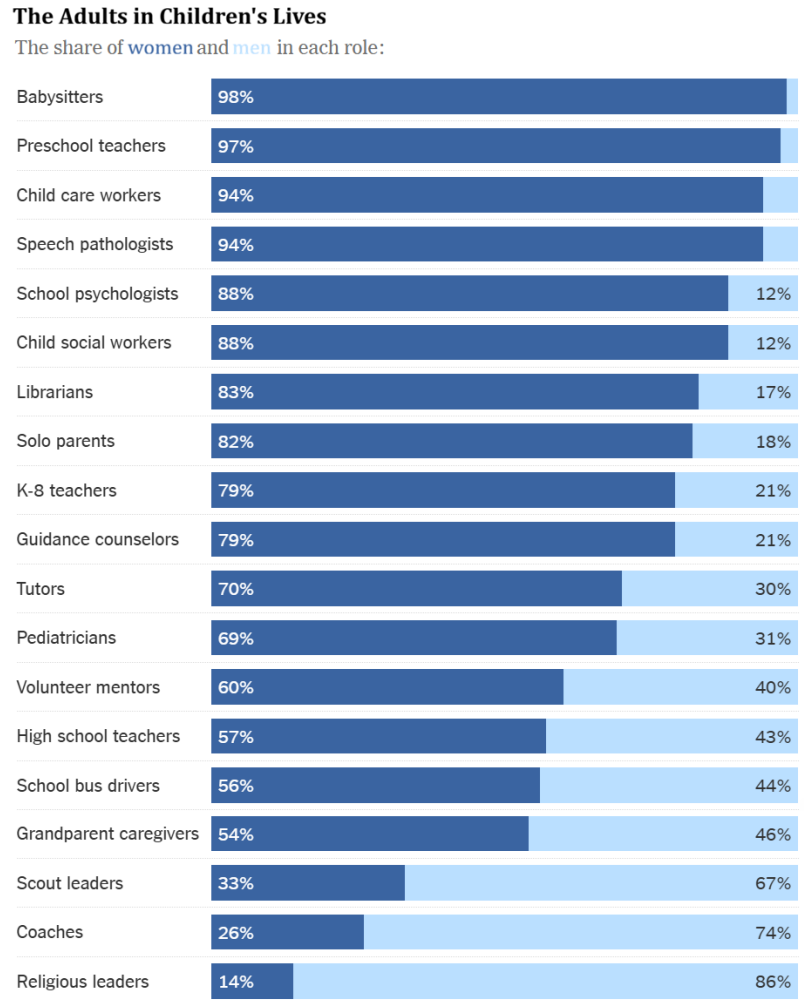
Also, homework 7 is posted and due on Sunday Mar. 29th

Let’s take a few minutes to explore the data visualizations you found as part of homework 5!

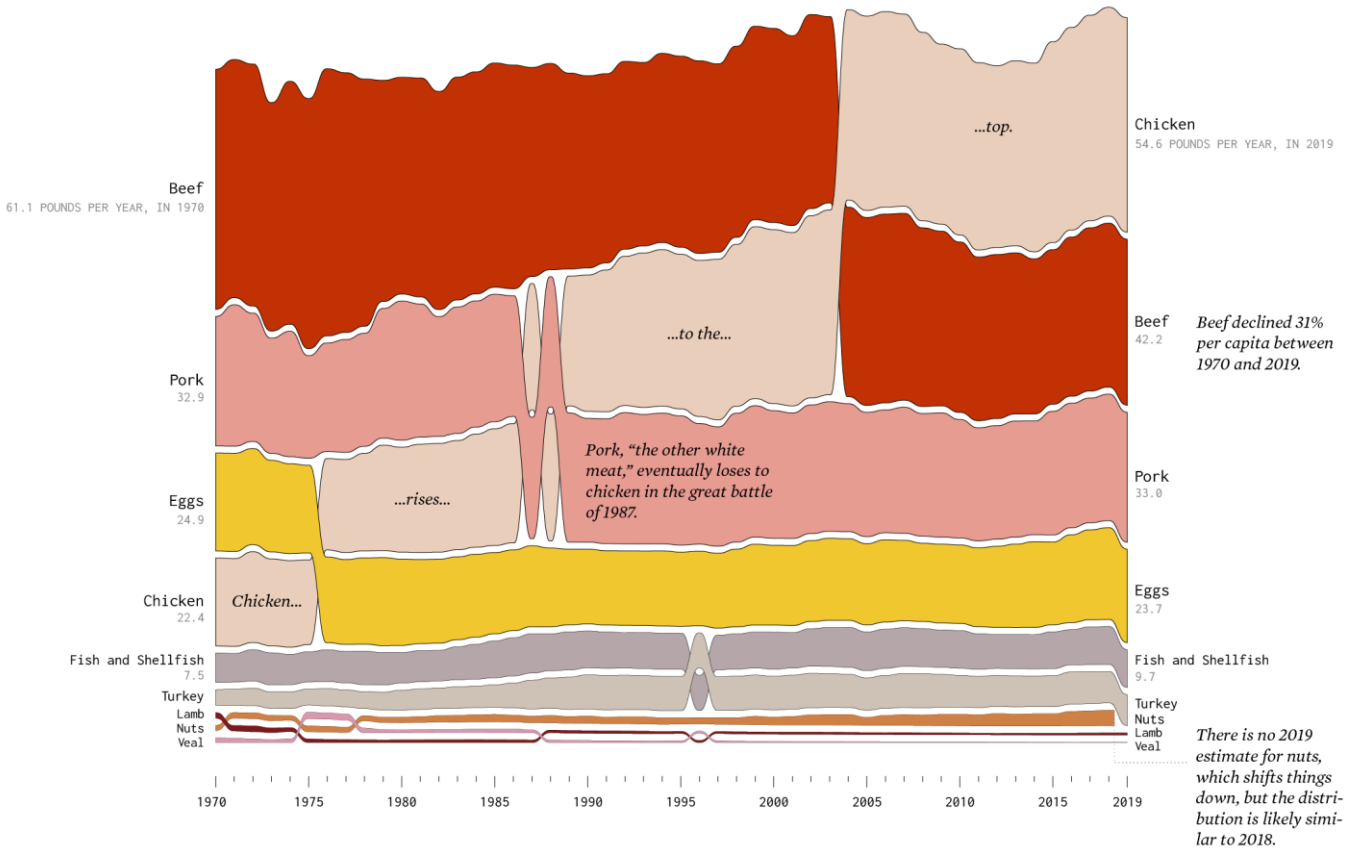
Bar charts

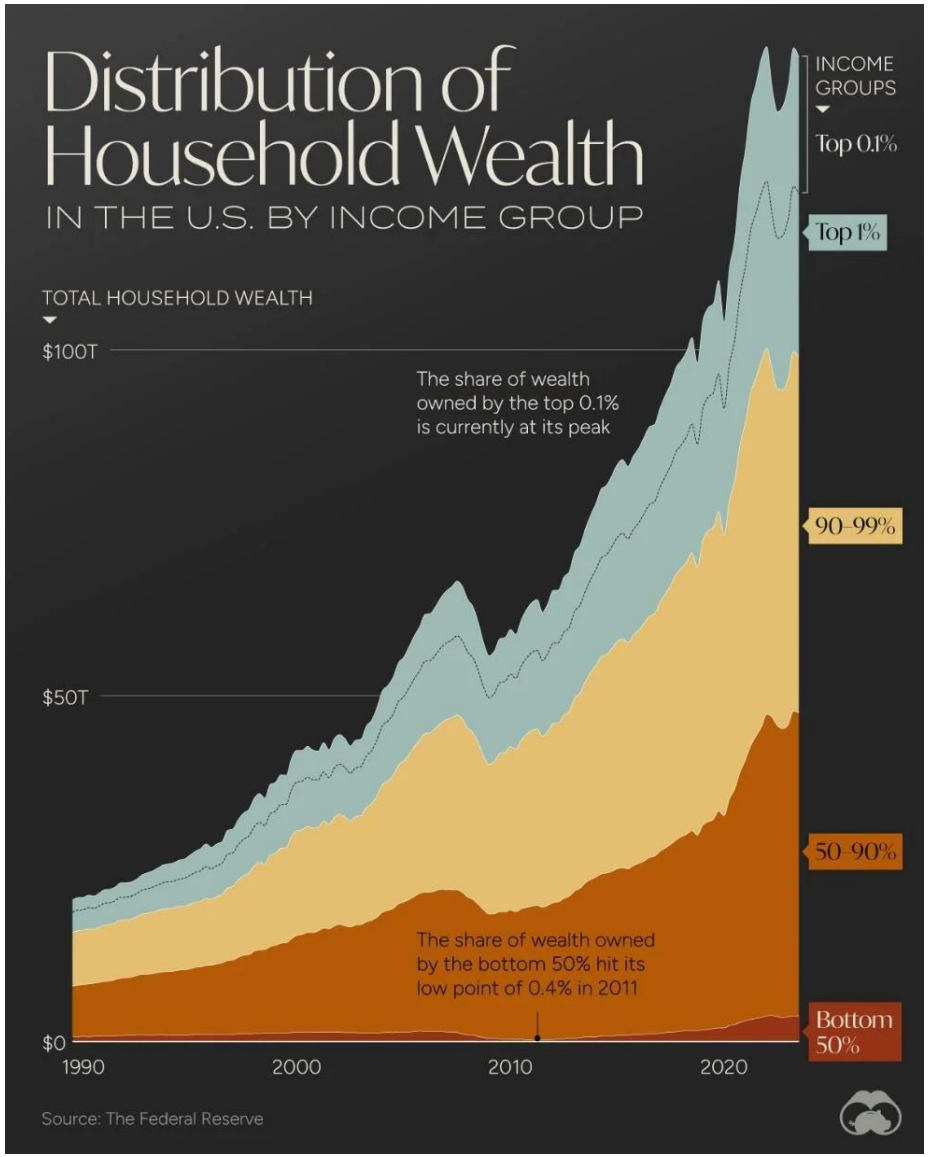

South Korea
Mid year 2024
Males
Male surplus
Females
Female surplus
100+
90
80
70
60
50
40
30
20
10
0
Age
600 000
400 000
200 000
0
200 000
400 000
600 000
Population
Source: Statistics Korea
Chart by: Kaj Tallungs


Social media reach & time spent among 15-24s
More 15-24s use Instagram and WhatsApp but they spend a lot more time on TikTok
Average Daily Reach & Ave Daily Mins PP among 15-24s
Daily Reach %
70%
60%
50%
40%
30%
20%
10%
0%
53%
52%
49%
45%
38%
23%
18%
8%
6%
29
18
22
88
118
25
23
8
10
40
20
0
15 PP
Instagram
WhatsApp
Facebook
Snapchat
TikTok
Twitter
Reddit
LinkedIn
Pinterest
15-24 Avg. Daily Mins PP
15-24 Avg. Daily Reach
Ipsos
Source: UKOM Ipsos iris Online Audience Measurement Service, August 2023
Social Media selected are at the Brand Group (BG) level from Ipsos iris. Facebook includes both Facebook and Messenger.

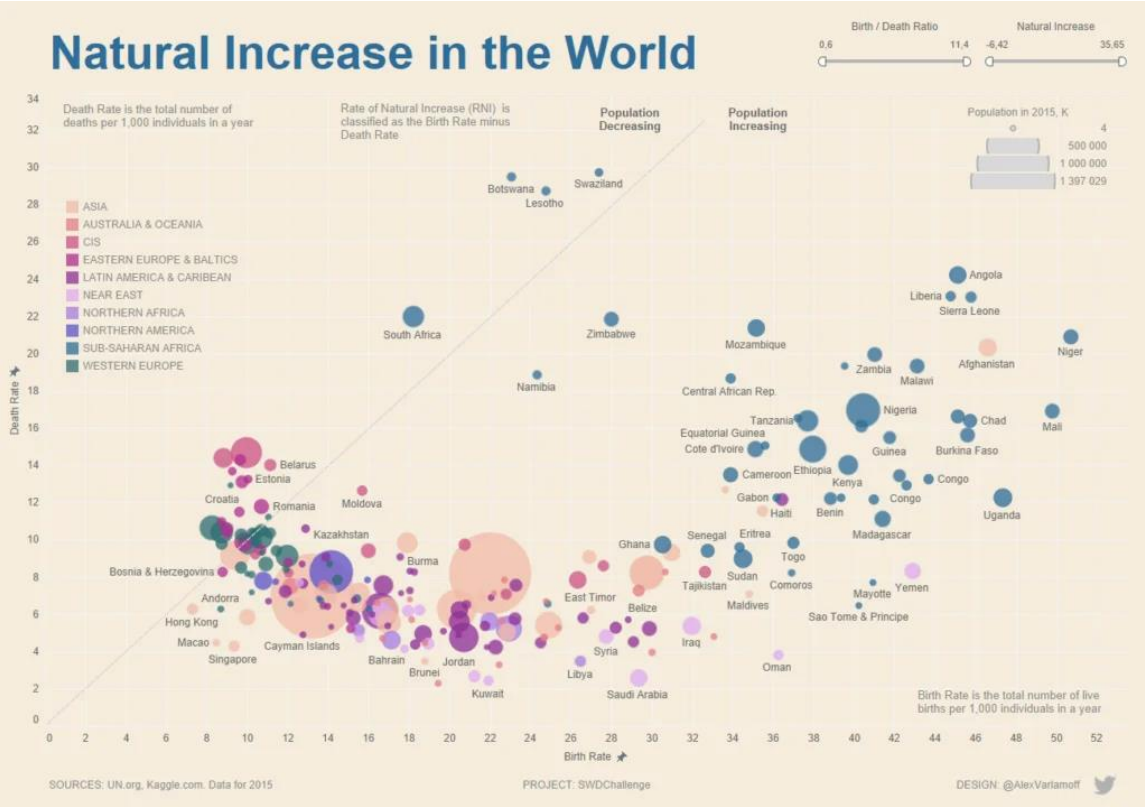

The Adults in Children's Lives
The share of women and men in each role:
Babysitters 98%
Preschool teachers 97%
Child care workers 94%
Speech pathologists 94%
School psychologists 88% 12%
Child social workers 88% 12%
Librarians 83% 17%
Solo parents 82% 18%
K-8 teachers 79% 21%
Guidance counselors 79% 21%
Tutors 70% 30%
Pediatricians 69% 31%
Volunteer mentors 60% 40%
High school teachers 57% 43%
School bus drivers 56% 44%
Grandparent caregivers 54% 46%
Scout leaders 33% 67%
Coaches 26% 74%
Religious leaders 14% 86%

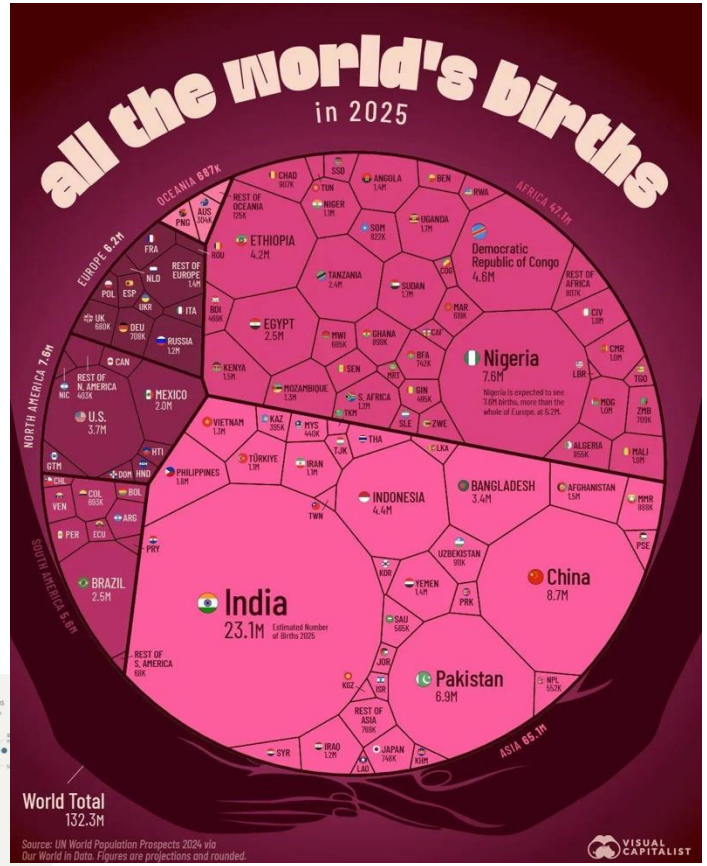
Time series

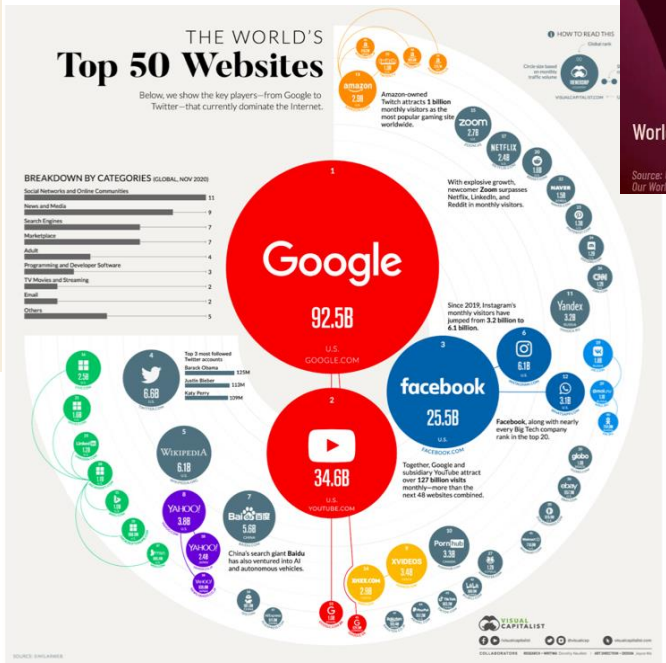

Beef
61.1 POUNDS PER YEAR, IN 1970
Pork
32.9
Eggs
24.9
Chicken
22.4
Fish and Shellfish
7.5
Turkey
Lamb
Nuts
Veal
...top.
...to the...
...rises...
Chicken...
Pork, "the other white meat," eventually loses to chicken in the great battle of 1987.
Chicken
54.6 POUNDS PER YEAR, IN 2019
Beef
42.2
Pork
33.0
Eggs
23.7
Fish and Shellfish
9.7
Turkey
Nuts
Lamb
Veal
Beef declined 31% per capita between 1970 and 2019.
There is no 2019 estimate for nuts, which shifts things down, but the distribution is likely similar to 2018.
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2019


Distribution of Household Wealth
IN THE U.S. BY INCOME GROUP
TOTAL HOUSEHOLD WEALTH
$100T
$50T
$0
The share of wealth owned by the top 0.1% is currently at its peak
The share of wealth owned by the bottom 50% hit its low point of 0.4% in 2011
INCOME GROUPS
Top 0.1%
Top 1%
90–99%
50–90%
Bottom 50%
1990
2000
2010
2020
Source: The Federal Reserve

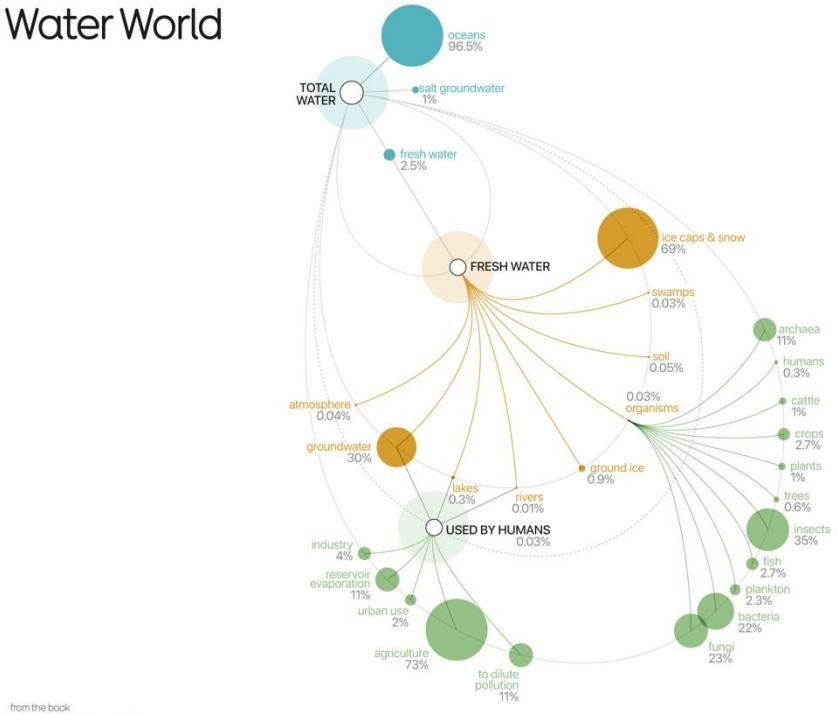
Visualizing different sizes

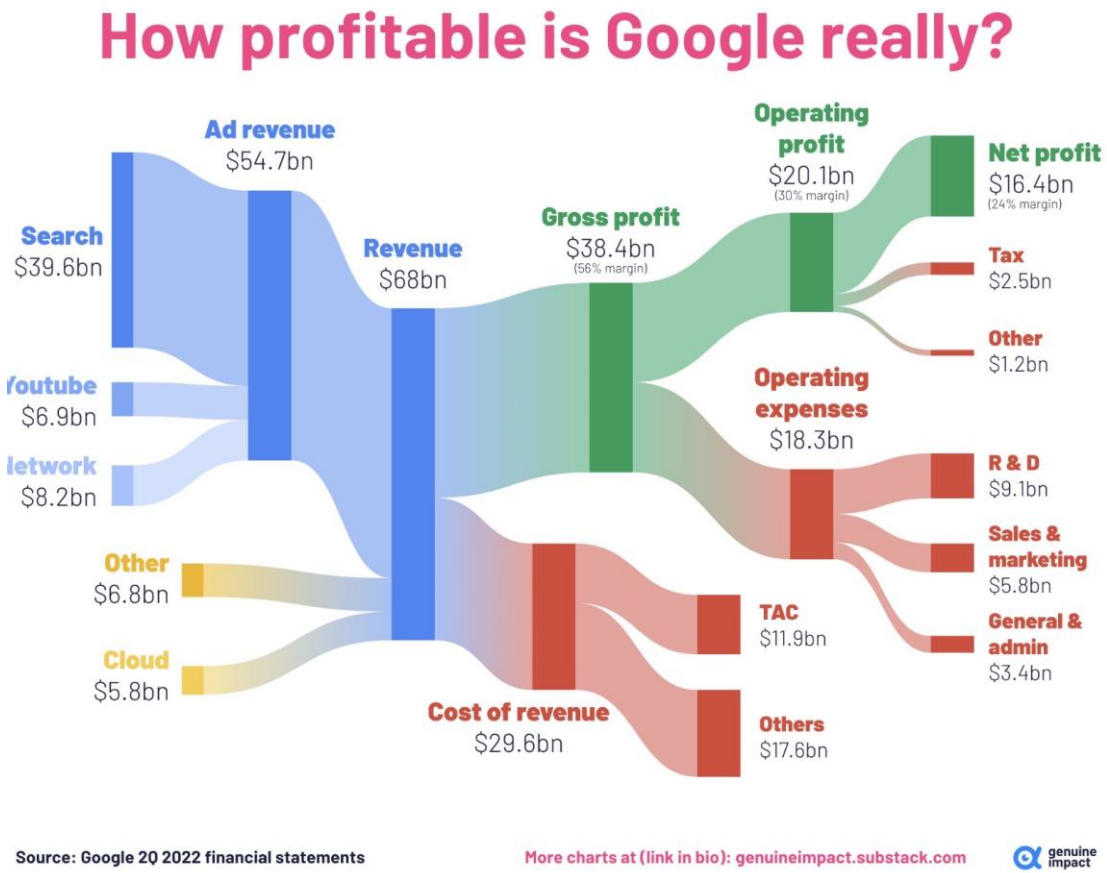




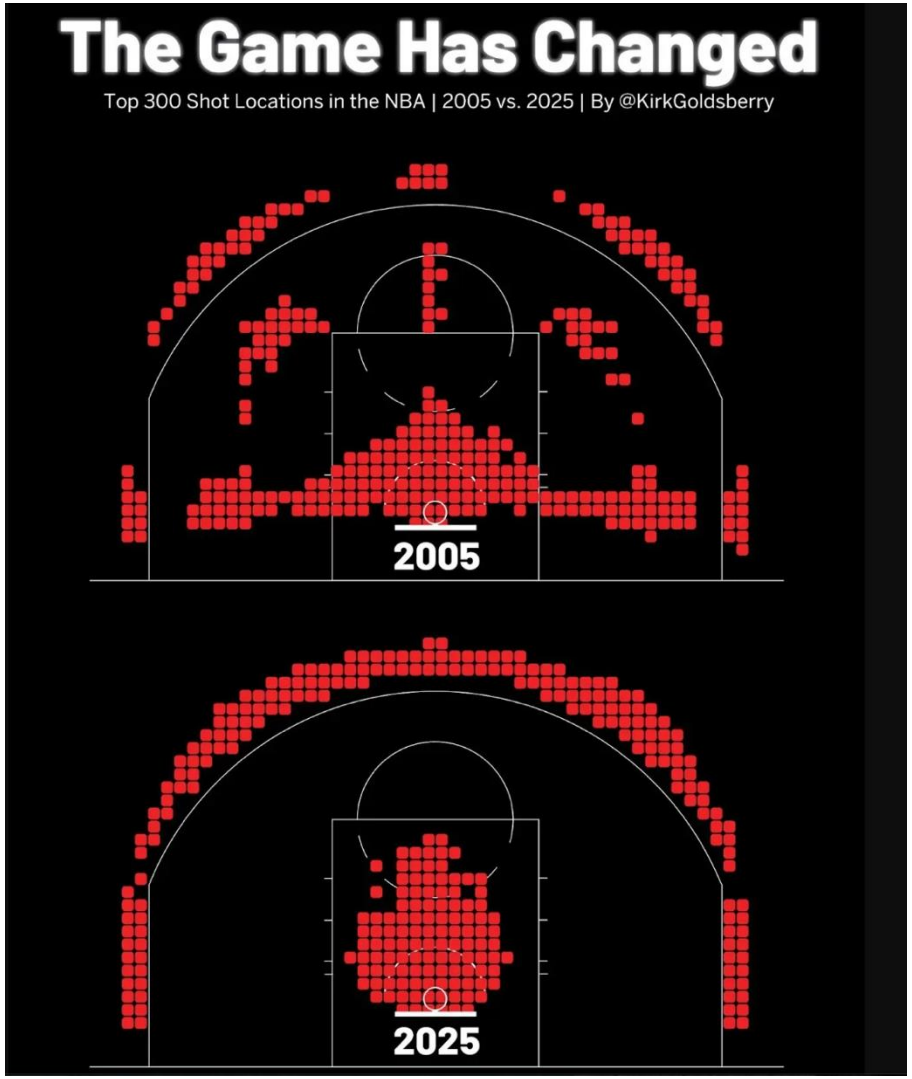


Information flow

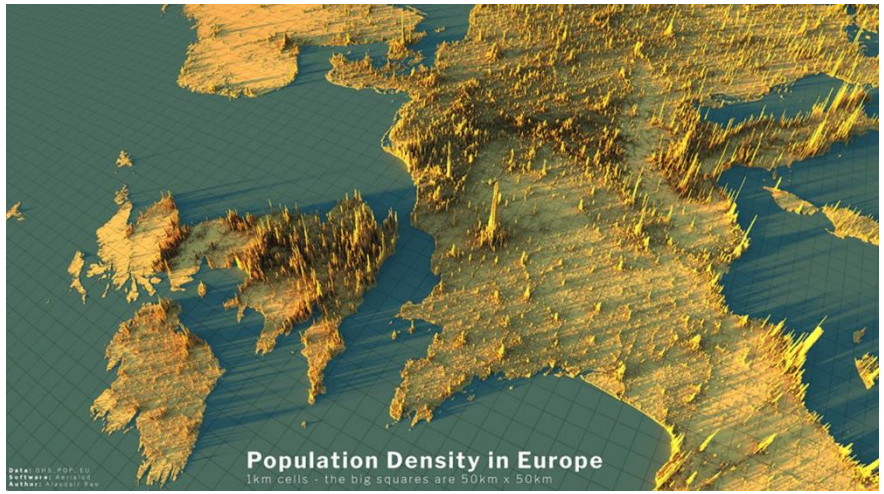


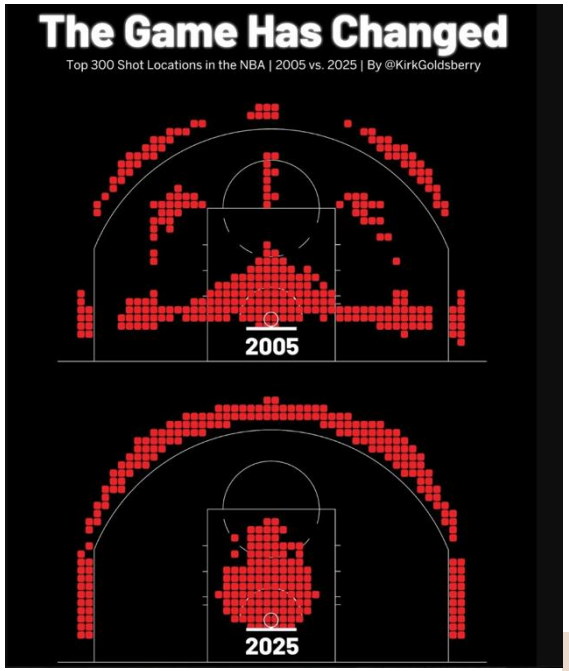


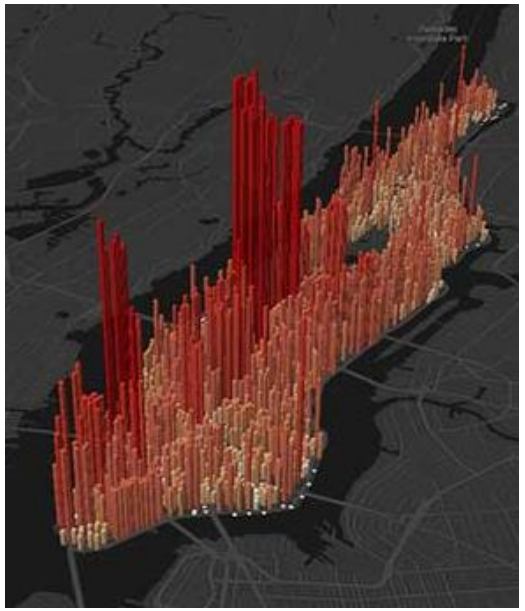
Spatial statistics

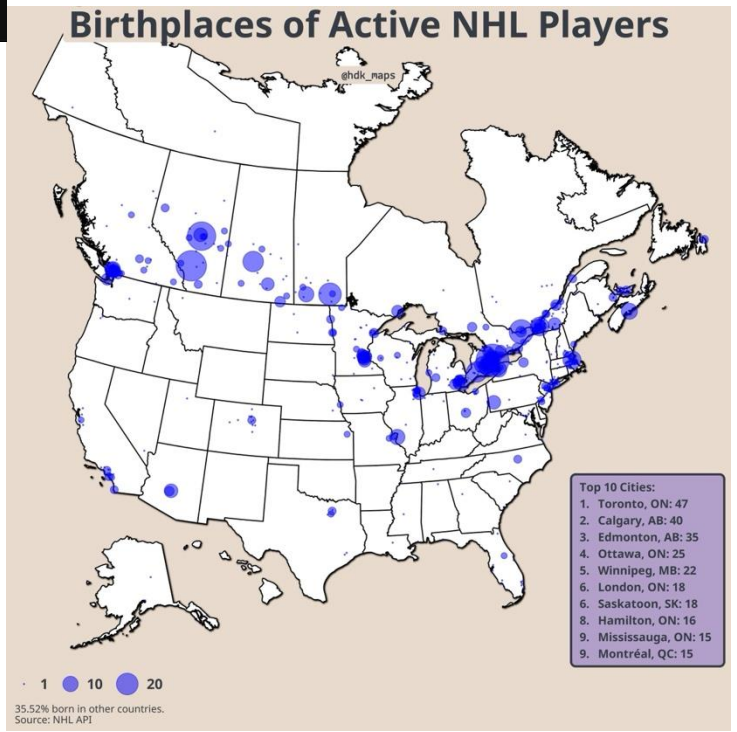


Maps

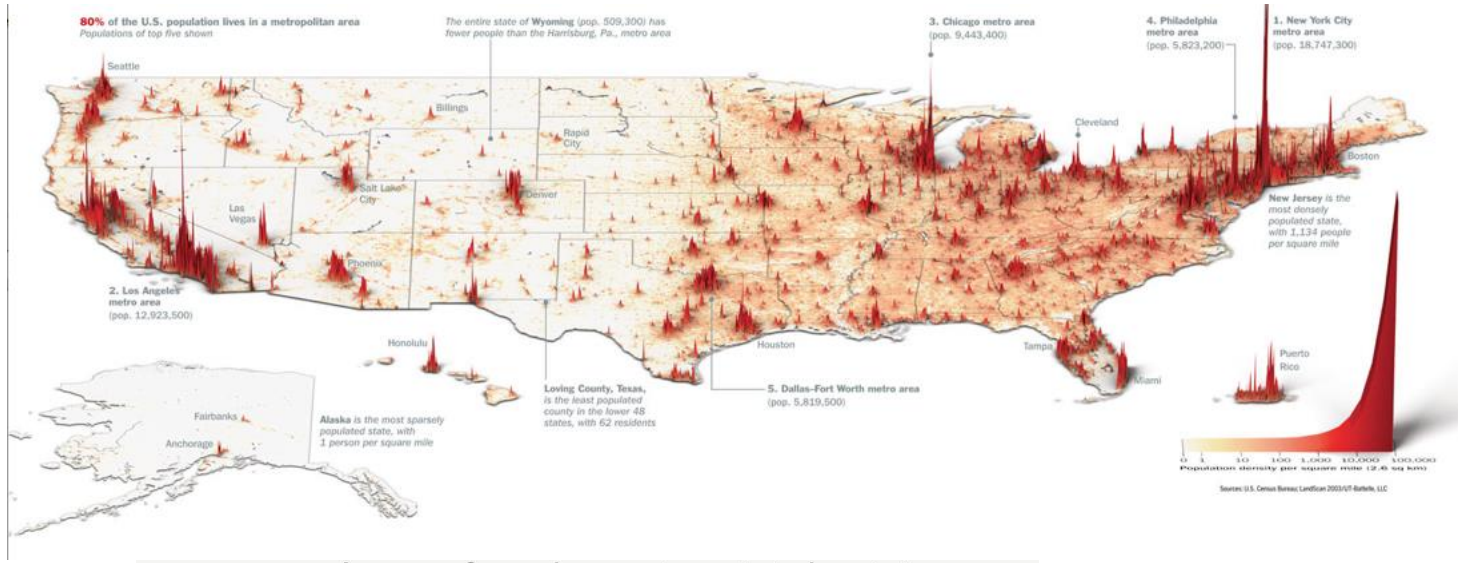

Population Density in Europe
1km cells - the big squares are 50km x 50km

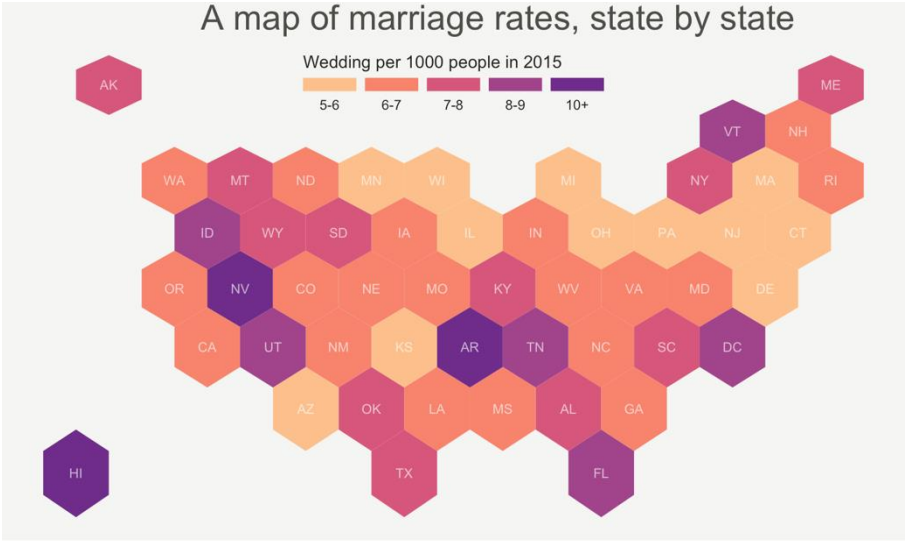

The Game Has Changed
Top 300 Shot Locations in the NBA | 2005 vs. 2025 | By @KirkGoldsberry
2005
2025

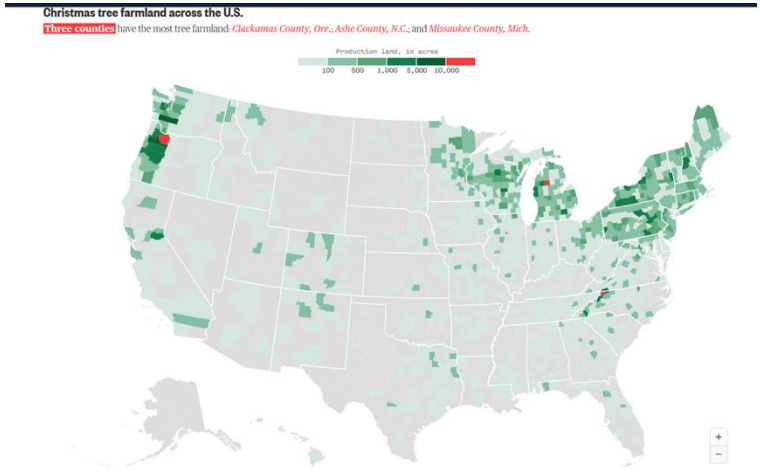



Birthplaces of Active NHL Players
@hdk_maps
Top 10 Cities:
1. Toronto, ON: 47
2. Calgary, AB: 40
3. Edmonton, AB: 35
4. Ottawa, ON: 25
5. Winnipeg, MB: 22
6. London, ON: 18
6. Saskatoon, SK: 18
8. Hamilton, ON: 16
9. Mississauga, ON: 15
9. Montréal, QC: 15
1
10
20
35.52% born in other countries.
Source: NHL API

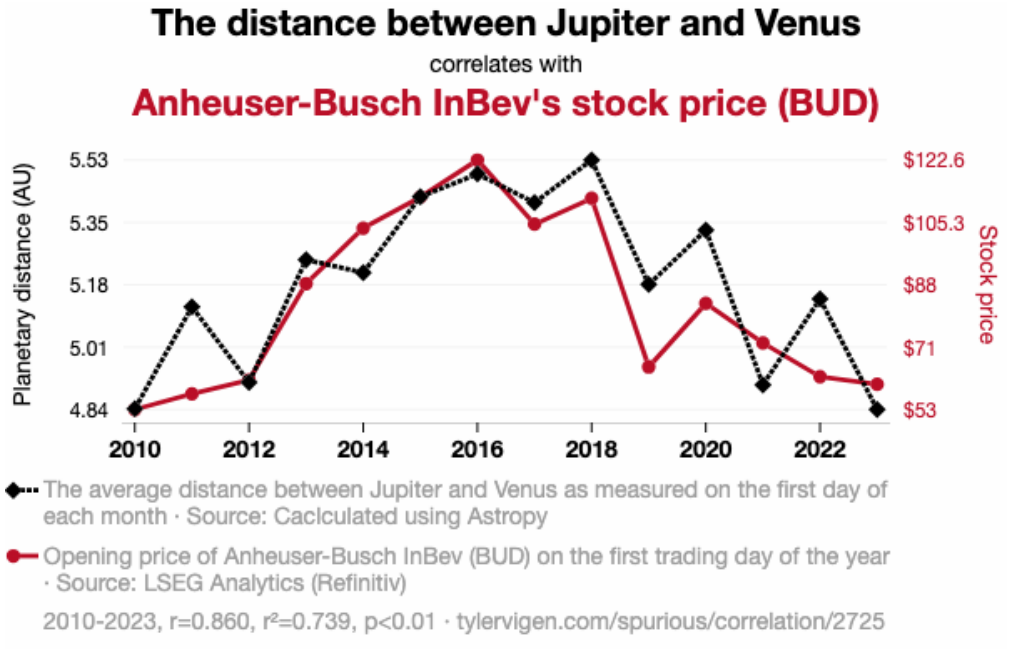
We love plotting on the US map

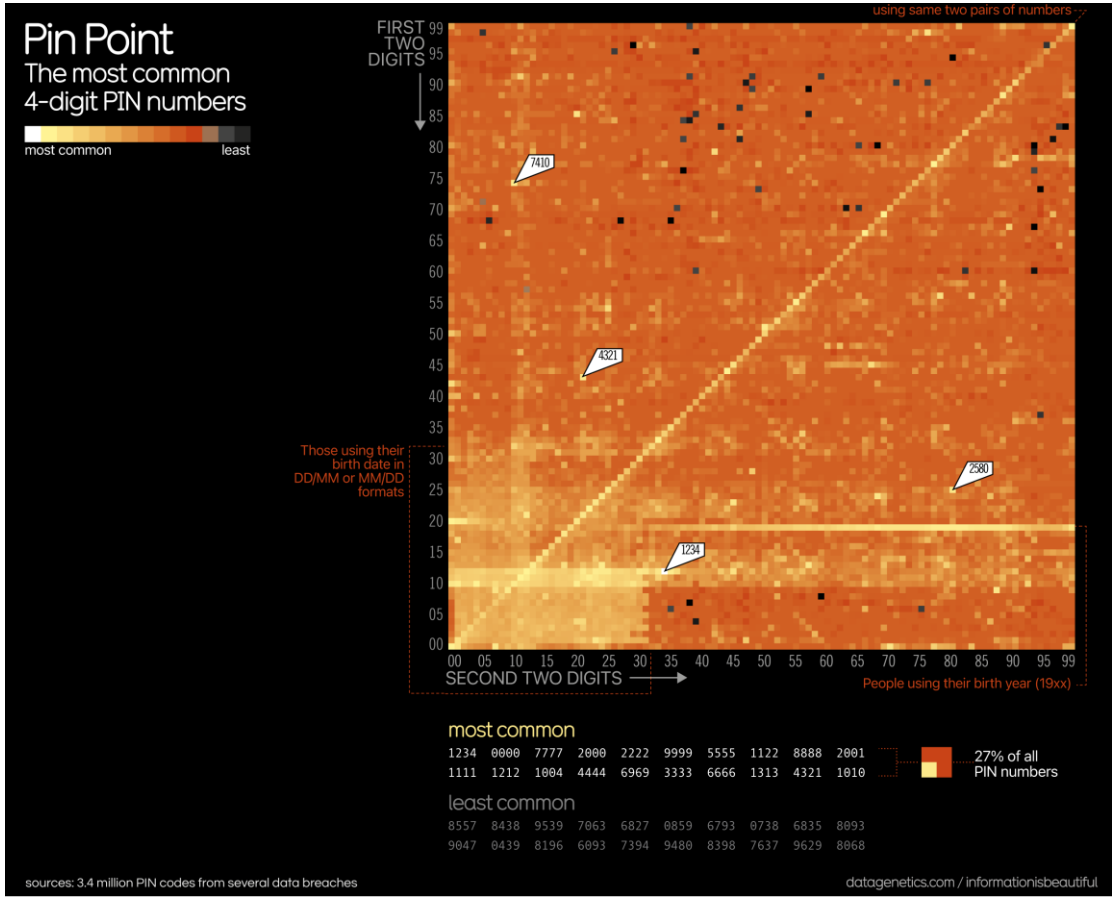






Interesting discoveries





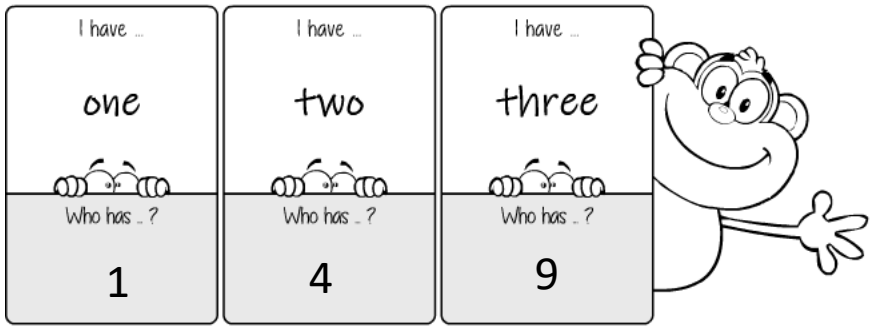
Quick review of for loops

Review: for loops

For loops repeat a process many times, iterating over a sequence of items

- Often we are iterating over an array of sequential numbers

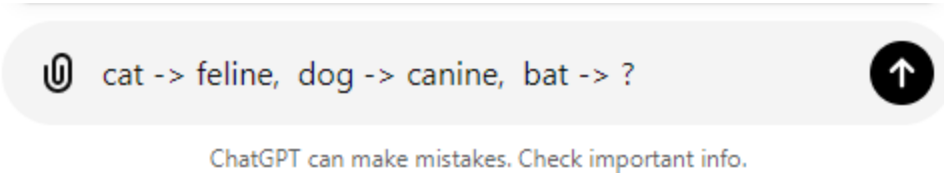
```
animals =  ["cat", "dog", "bat”]
for creature in animals:
    print(creature)

for i in range(10):
    print(i**2)
```



Review: enumerate and zip

We can use the enumerate() function to both items in a list, and sequential integers:

```
animals =  ["cat", "dog", "bat”]
        for i, creature in enumerate(animals):
                print(i, creature)
```



We can use the zip() function to get items for two lists:

```
        animal_order =  ["feline", "canine", "chiropteran"]
        for curr_order, curr_animal in zip(animal_order, animals):
                print(curr_order, curr_animal)
```

Let’s go over an example in Jupyter!

Conditional statements

Review: comparisons

We can use mathematical operators to compare numbers and strings

- Results return Boolean values True and False

Comparison	Operator	True example	False Example
Less than	<	2 < 3	2 < 2
Greater than	>	3 > 2	3 > 3
Less than or equal	<=	2 <= 2	3 <= 2
Greater or equal	>=	3 >= 3	2 >= 3
Equal	==	3 == 3	3 == 2
Not equal	!=	3 != 2	2 != 2

We can also make comparisons across elements in an array

Conditional statements

Conditional statements control the sequence of computations that are performed in a program

We use the keyword if to begin a conditional statement to only execute lines of code if a particular condition is met.

We can use elif to test additional conditions

We can use an else statement to run code if none of the if or elif conditions have been met.

```
num = 5
if num == 1:
    print("Monday")
elif num == 2:
    print("Tuesday")
elif num == 3:
    print("Wednesday")
elif num == 4:
    print("Thursday")
elif num == 5:
    print("Friday")
elif num == 6:
    print("Saturday")
elif num == 7:
    print("Sunday")
else:
    print("Invalid input")
```

Let's explore this in Jupyter!

Defining functions

Writing functions

We have already used many functions that are built into Python or are imported from different modules/packages.

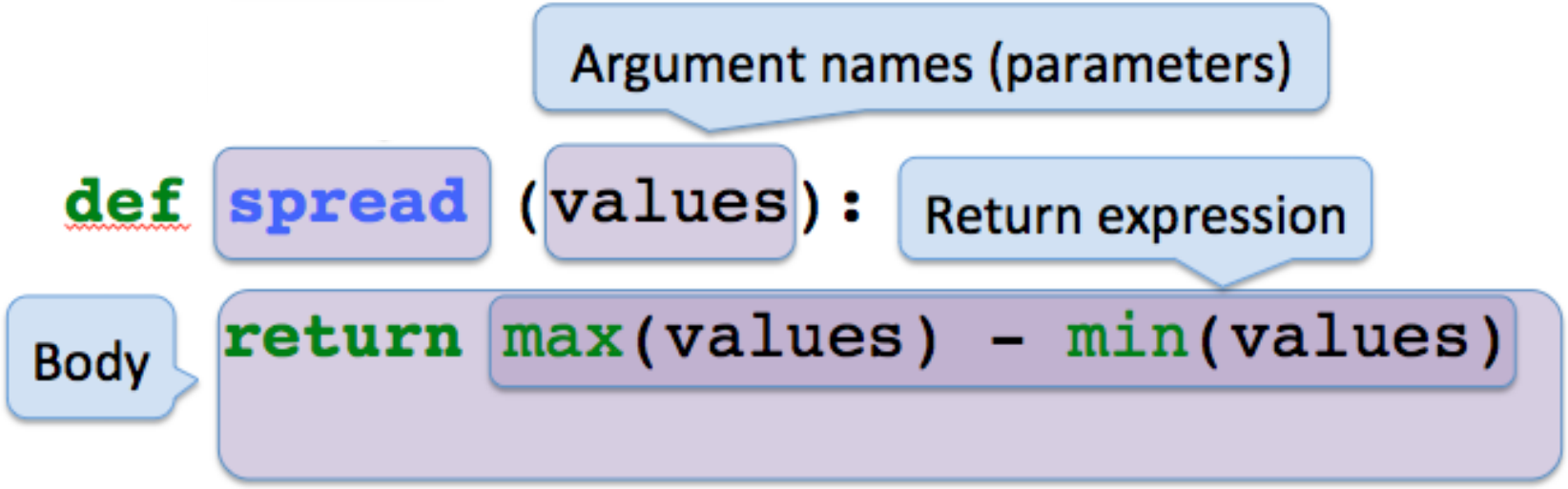
Examples…???

- sum()
- statistics.mean()
- np.diff()
- etc.

Let’s now write our own functions!

Def statements

User-defined functions give names to blocks of code



Let's explore this in Jupyter!

Practice: simulating flipping coins

Simulating flipping a coin

Let's practice writing functions by writing a function that can simulate flipping coins, where each coin has **π** probability of being heads

- Where **π** is a number between 0 and 1; e.g., **π** = 0.5 is a fair coin

We can do this using the following procedure:

1. Generate a random number between 0 and 1

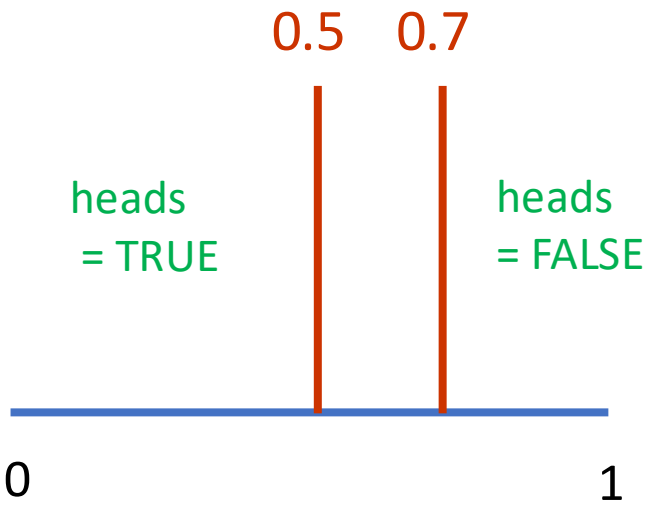
```
rand_num = np.random.rand(1)
```

2. Simulate a fair coin (.5) by mark values less than .5 as heads (True)

```
heads = rand_num <=  .5
```

3. We can simulate a biased coin that will come up with heads 70% of the time (**π** = 0.7) using:

```
rand_num = np.random.rand(1)
heads = rand_num <=  .7
```



Simulating n random coin flips

We can simulate the number of heads we would get flipping a coin ***n*** times using:

1. Generate ***n*** random numbers uniformly distributed between 0 and 1

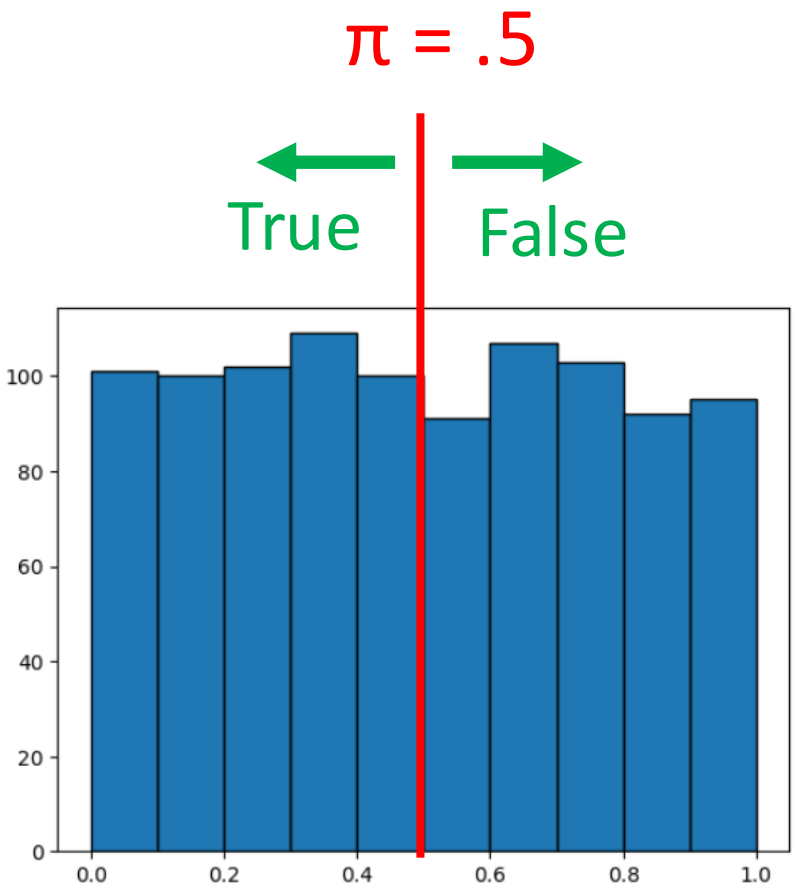
```
rand_nums = np.random.rand(n)
```

2. Mark points less than π as being True, and greater π than as being False

```
rand_binary = rand_nums <=  prob_value
```

3. Sum the number of heads (True's) we get

```
num_heads = np.sum(rand_binary)
```



Let's explore this in Jupyter!